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Rising Replacement Burden and the Compression of Net Building

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Abstract

The share of output absorbed by replacing depreciating productive structure — the ratio of the consumption of fixed capital to output, $\delta = \text{CFC}/Y$ — has risen steadily across every major economy since 1950. We establish three things. First, the rise is a robust and physical fact: it holds across US national accounts (1929–2025), a 180-country Penn World Table panel (1950–2023), and OECD STAN industry data (1970–2023); it survives a purely real construction that removes all nominal prices; and it is invariant to the depreciation-accounting method national agencies use. Second, the rise tightens an accounting constraint. Because the gross investment share is bounded (Kaldor 1961), a rising δ compresses the residual available for net new building, $n = C/Y - \delta$; in Japan this residual has fallen to 2.5% of GDP, the closest empirical approach to a *replacement trap* in which replacement absorbs all gross investment and net building reaches zero. Third, the rise decomposes — through the identity $\delta = (\text{CFC}/K) \times (K/Y)$ — into a composition channel (faster-depreciating capital) and a scale channel (deeper accumulation); the United States is composition-led and Japan scale-led, with the intermediate cases sensitive to valuation and best read as qualitative. We stop at the net building rate. Whether the compressed n slows productivity growth runs through a transmission coefficient whose magnitude is debated in the growth-accounting literature and which we do not identify; we establish the measurement and the identity, and leave the causal step to subsequent work.

1. Introduction

Labor productivity growth in the advanced economies has been declining for seven decades. In the United States it fell from 3.6% per year in the 1950s to 0.8% in the 2020s, and the pattern is universal: every major industrialized economy has decelerated, differing only in timing. The literature has named many proximate causes — the exhaustion of transformative innovations (Gordon 2016), declining research productivity (Bloom, Jones, Van Reenen, and Webb 2020), energy shocks (Hamilton 2009), financialization (Cecchetti and Kharroubi 2012), secular stagnation (Summers 2014), the fading of IT gains (Fernald 2015).

This paper documents a prior, mechanical fact that sits underneath those explanations rather than competing with them. Every economy that accumulates productive structure must devote a growing share of its output to maintaining what it has already built. As the stock grows, so does the replacement burden; and because the share of output going to gross investment is bounded, a rising replacement burden leaves an ever-smaller residual for net new building. This is an accounting constraint, and we show that its binding has tightened in lockstep with the collapse of net capital formation across every economy and data layer we examine. Japan, at 2.5% of GDP in net capital formation, is the closest empirical approach to the limiting case in which replacement absorbs all of gross investment.

The contribution is a measurement and an identity, and we are deliberate about where it stops. The chain from a rising δ to a compressed net building rate $n = C/Y - \delta$ is arithmetic and requires no behavioral assumption. The further chain — from a com-

pressed n to slower productivity growth — runs through the relationship between net accumulation and productivity (Solow 1956; Jorgenson, Ho, and Stiroh 2005), whose structural magnitude is itself debated, and which in observational data is difficult to separate from reverse causation and from movements in the shared output denominator. We do not estimate that transmission or assert its sign as a causal matter, and our own low-frequency tests cannot identify it (Section 5.3). What we establish is upstream of it and does not depend on it.

The analysis proceeds in three layers, each a distinct source: the United States (BEA and BLS, 1929–2025) for depth and asset-level decomposition; a Penn World Table 11.0 panel (1950–2023) for breadth, establishing that the rise is not a feature of any one country or accounting convention; and OECD STAN (1970–2023) for sectoral granularity and the net capital stock that drives the decomposition. The three draw on overlapping primary records, so agreement across them is evidence on robustness to processing choices rather than three independent measurements. Section 2 develops the accounting framework and states precisely what is and is not claimed. Sections 4–6 present the three layers. Section 7 examines Japan as the limiting case. Sections 8–9 discuss and conclude. Appendix E collects the apparatus toward causal identification, which is not part of this paper’s claims.

2. Accounting Framework

2.1 An identity

Let $\Omega(t)$ denote the aggregate stock of organized productive structure. Its rate of change equals inflows minus outflows:

$$\dot{\Omega}(t) = C(t) - D(t), \quad (1)$$

where $C(t)$ is the construction flow and $D(t)$ the replacement flow. This is an accounting identity. We measure D by the Consumption of Fixed Capital (CFC) — a lower bound on the true replacement flow, since CFC excludes institutional maintenance, administrative overhead, and regulatory compliance — and C by Gross Fixed Capital Formation (GFCF). Define

$$\delta = D/Y, \quad \rho = D/C, \quad n = C/Y - \delta : \quad (2)$$

the replacement share of output, the replacement share of gross investment, and the net building rate. The stagnation condition $\rho = 1$ means all investment is replacement and $\dot{\Omega} = 0$.

2.2 From identity to a compressed residual

One restriction moves the identity to a statement about trends.

Assumption S. $D(t) = \mathcal{D}(\Omega(t))$, with $\mathcal{D}$ increasing: a larger stock requires a larger replacement flow. This is supported by engineering estimates (National Research

Council 2012) and by the positive cross-country correlation of CFC with the net capital stock.

Given Assumption S and a bounded gross investment share (Kaldor 1961; Jones 2016), the net building rate $n = C/Y - \delta$ is compressed as δ trends upward. This is identity. Because δ moves slowly relative to output (Appendix A.5), it is a slow variable on a decadal timescale; as $\rho \rightarrow 1$, net building approaches zero — the replacement trap. Three frictions keep the market from arresting the rise: much high-maintenance stock has public-good character; the shift toward short-lived assets is individually rational but collectively raises aggregate δ , a composition externality (Acemoglu, Aghion, Bursztyn, and Hemous 2012); and the savings rate is bounded by consumption smoothing.

The decomposition. The identity $\delta = D/Y$ factors as

$$\delta = \underbrace{(D/\Omega)}_{\text{effective depreciation rate}} \times \underbrace{(\Omega/Y)}_{\text{capital-output ratio}},$$

so a rising δ can reflect either a faster-depreciating stock (the **composition channel**, D/Ω up) or deeper accumulation (the **scale channel**, Ω/Y up). Both compress n by the same arithmetic; the distinction is interpretive, locating the relevant margin in the *mix* of capital versus its *quantity*. We estimate the decomposition country by country in Section 5.3.

2.3 Relationship to Solow (1956), stated narrowly

In the textbook Solow model the depreciation rate is an exogenous constant. The fact we document — that the effective depreciation rate CFC/K rises systematically with the maturity and composition of the capital stock — is inconsistent with that *modeling choice*, not with the growth framework, which accommodates a rising endogenous δ once the assumption is relaxed. We falsify the constant- δ assumption, nothing larger.

2.4 Testable accounting implications

The paper’s claims are measurement and identity. The framework implies four testable regularities:

- **I1.** δ is highly persistent — a slow variable. (*Requires nothing; tested in §4.1.*)
- **I2.** $\rho = D/C$ trends toward 1 in maturing economies. (*Requires a bounded C/Y ; §6.2.*)
- **I3.** The rise of δ decomposes into a composition channel and a scale channel, both operative. (*The multiplicative identity; §4.3, §5.3.*)
- **I4.** As $\rho \rightarrow 1$, $n \rightarrow 0$. (*Follows from (1) and I2; the Japan limiting case, §7.1.*)

Each is falsifiable: the framework fails if δ is not rising, if ρ does not trend toward 1 in mature economies, or if the decomposition shows neither channel operating. The framework makes no claim, here, about any causal effect of δ on productivity growth, its sign, or its magnitude; that question is deferred (§8.2).

3. Data

The three layers draw on overlapping primary records — national-agency CFC estimates — processed under different aggregation rules and depreciation assumptions. They are independent at the level of construction, not of source data.

Layer 1 — United States (NIPA, 1929-2025). BEA NIPA and BLS multifactor productivity. $\delta = \text{CFC}/\text{GDP}$, $\rho = \text{CFC}/\text{GFCF}$, net investment rate. BEA Fixed Asset Accounts provide current-cost net capital stock and depreciation by asset type, enabling $\delta = (\text{CFC}/K) \times (K/Y)$.

Layer 2 — Cross-country breadth (PWT 11.0, 1950-2023). Penn World Table 11.0 (Feenstra, Inklaar, and Timmer 2015), used to establish breadth and a purely real construction of δ (§5.2). Not used for the composition/scale decomposition.

Layer 3 — OECD STAN (2025 edition, 1970-2023). ISIC Rev. 4; $\delta = \text{CFC}/\text{VA}$, ρ , net investment/VA, and net capital stock (measure N11NA, current replacement cost), valued consistently with the CFC flow. Total-economy net stock is available for 33 countries; long series (≥ 44 years) for 13.

4. Layer 1: The United States

4.1 The rise of δ , and its persistence

Figure 1: The D/C ratio and the compression of net building (US NIPA, 1929-2025)

The US D/C ratio rose from about 0.04 in the 1940s to 0.10 in the 2020s (post-war trend $R^2 = 0.864$); maintenance as a share of GDP rose from 3.5% in 1944 to 8.4% in 2025. Gross investment composition has flipped: replacement's share of gross investment rose from 39% in the 1950s to 59% in the 2020s, while net new building fell from 61% to 41%. The series is highly persistent — significant autocorrelation through lag 9, a memory far exceeding any business-cycle frequency (I1).

4.2 Co-movement with productivity

MFP growth and δ trace near-mirror images over seven decades, with a low-frequency (HP-filtered) trend correlation of $r = -0.807$. We report this to motivate why the constraint is of interest. We do not read it as a causal channel: both series are highly persistent and share the output denominator, and the direction of any structural relationship cannot be settled by the correlation. The instruments, natural experiments, and frequency analysis that a causal argument would require are collected in Appendix E.

4.3 The US rise is composition-led

The decomposition $\delta = (\text{CFC}/K) \times (K/Y)$ on BEA Fixed Asset Accounts shows the US rise running through composition. The effective depreciation rate for private nonresidential capital rose from 5.7% in 1950 to 8.9% in 2024 (trend $R^2 = 0.93$) while

D/C Ratio Analysis (US NIPA, 1929-2025)
D = Consumption of Fixed Capital, C = GDP – CFC

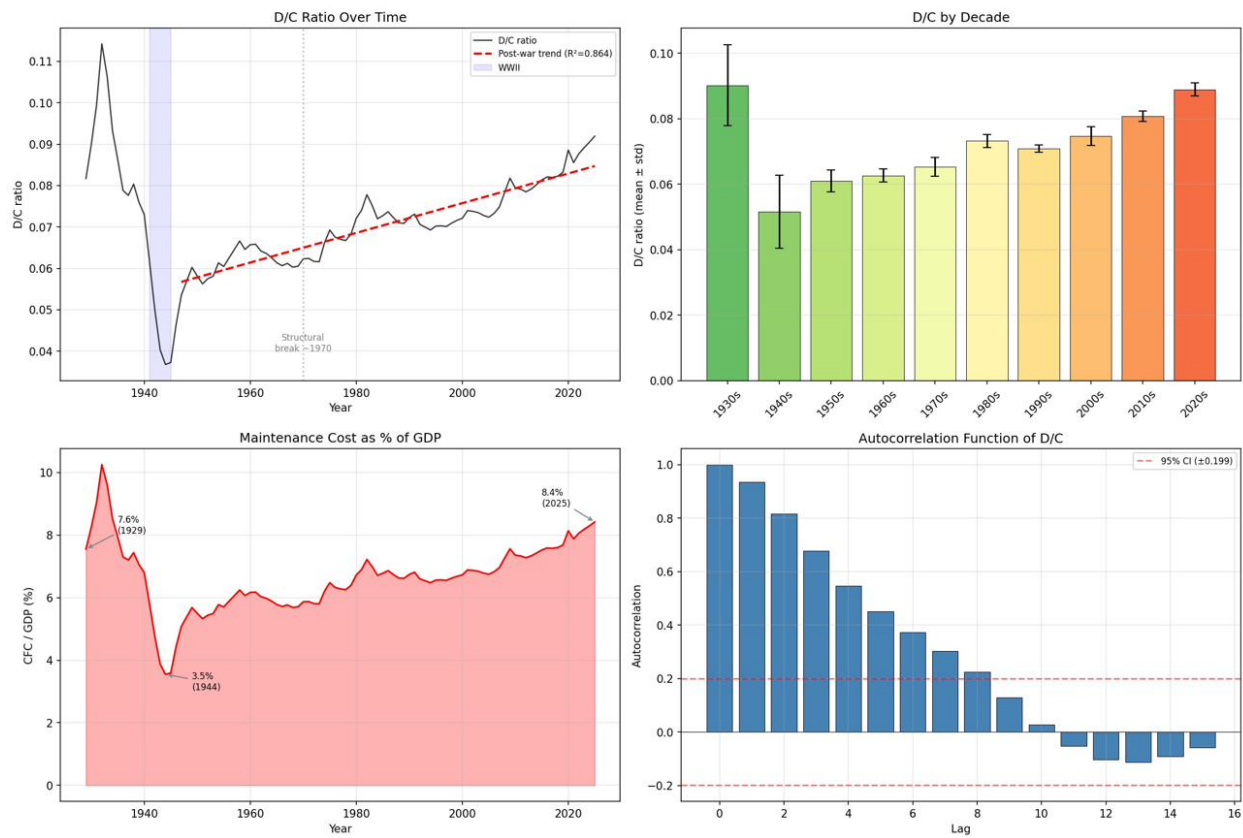


Figure 1: DC and net building

the capital-output ratio was flat to falling. No individual asset class depreciates faster: equipment held at 12–14% and structures at 2.5–3% throughout. What changed is the mix. Intellectual property products — software, R&D, entertainment originals — depreciate at 14–24% per year, five to eight times faster than structures, and rose from 3.8% of the private nonresidential capital stock in 1950 to 15.1% in 2024. On the flow side, IPP rose from 9% of replacement to 42%. A growing share of each year’s replacement goes to assets that were themselves recently built.

Figure 2: Decomposition of the US rise in δ — composition versus scale

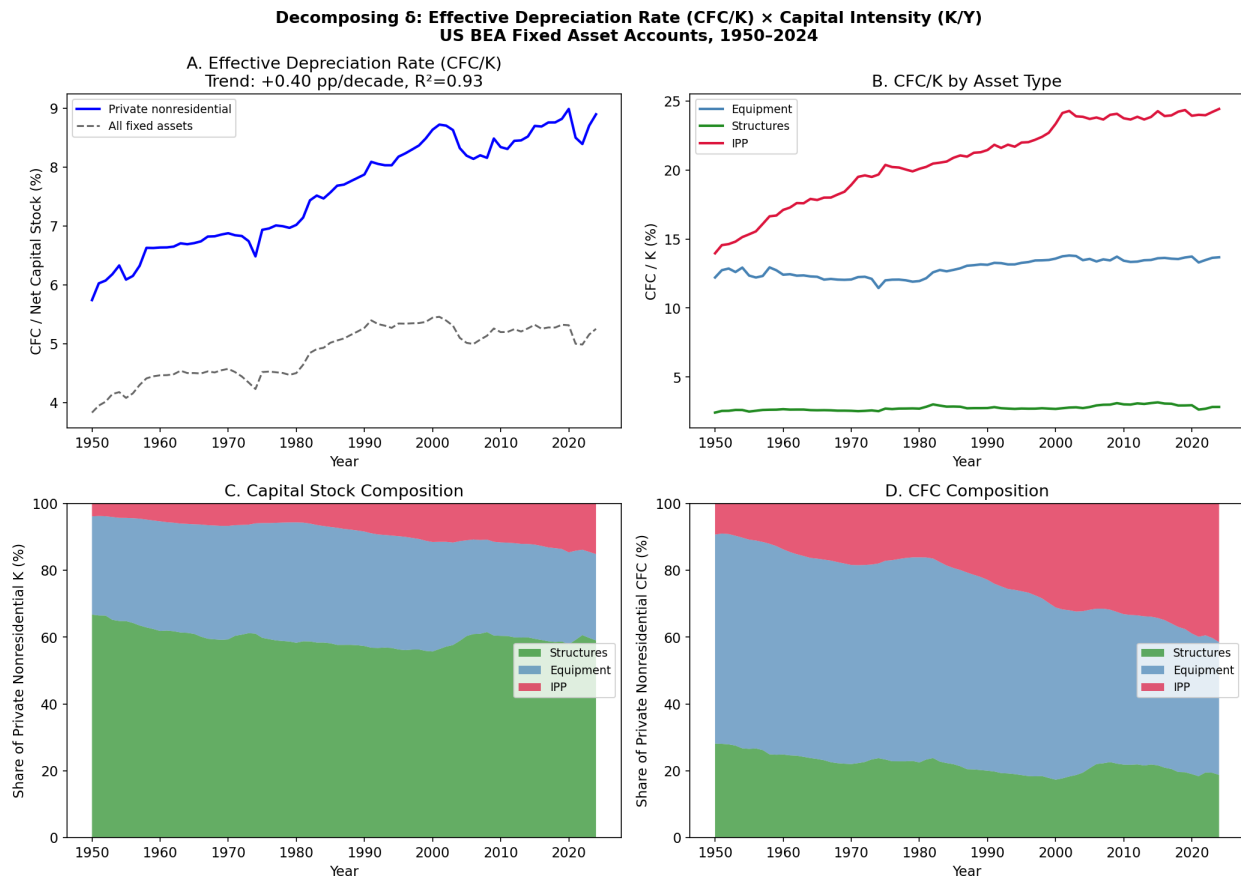


Figure 2: US decomposition

4.4 The named slowdown sources are sources of δ

The structural changes the slowdown literature has named — shifts toward short-lived assets, aging infrastructure, financialization, the R&D paradox — each raise δ through a different sub-component, and each is positively associated with δ in US data, consistent with their being *upstream* of the replacement burden rather than independent of it. The R&D productivity paradox (Bloom et al. 2020) reads naturally on this account: R&D creates fast-depreciating IPP ($\approx 31.5\%/yr$), so rising R&D mechanically raises δ , and part of its measured productivity benefit is offset by the replacement cost of what it creates. We offer this as a consistent reinterpretation,

not a causal decomposition; the underlying regressions are productivity-on- δ objects and are reported in Appendix E.

5. Layer 2: Cross-Country Evidence

5.1 δ rises everywhere

Figure 3: The rise of δ across 180 countries

The rise is not a US phenomenon. In the 28-country core panel, δ trends upward in every economy, with slopes from 0.46 pp/decade (US) to 3.20 pp/decade (South Korea), across all income groups. In $(\delta, \text{productivity})$ space every economy travels from the upper left toward the lower right; the speed differs but the direction is universal.

5.2 The rise is physical, not a price artifact

If the relative price of investment goods were falling, nominal CFC/GDP could rise with an unchanged physical burden. Constructed purely from real (volume) PWT capital stocks, with no nominal prices at any stage, real δ rises in 26 of 28 core countries, and the real trend is *steeper* than the nominal in 22 of 28 — the falling relative price of ICT capital means nominal CFC understates the physical expansion. The rise is physical. It is also invariant to accounting method: classifying countries by depreciation function (geometric, linear, hyperbolic; Pionnier, Zinni, and Baret 2023), the rise is present under every method, and the cross-method interaction is statistically indistinguishable from zero ($p = 0.949$).

5.3 Decomposing the rise across countries

We decompose the rise at the total-economy level using OECD STAN net capital stock (N11NA, current replacement cost), so numerator and denominator share a valuation basis with the CFC flow; the identity $\delta = (\text{CFC}/K) \times (K/Y)$ holds to machine precision. For each country we compare the first and last decades of available data and attribute $\Delta\delta$ to the two channels by a mean-value decomposition.

Table 1: Decomposition of rising δ , first decade versus last (OECD STAN, current replacement cost)

Country	Span	$\Delta\delta$	via CFC/K	via K/Y	Leading channel
United States	1975–2021	+0.029	+0.038	−0.009	composition
Austria	1976–2023	+0.034	+0.042	−0.009	composition
Belgium	1975–2023	+0.040	+0.045	−0.005	composition

Country	Span	$\Delta\delta$	via CFC/ K	via K/Y	Leading channel
Denmark	1975–2023	+0.033	+0.045	−0.012	composition
Norway	1975–2023	+0.011	+0.013	−0.001	composition
Australia	1975–2023	+0.030	+0.022	+0.008	composition
Finland	1975–2023	+0.045	+0.025	+0.020	mixed
Italy	1980–2023	+0.048	+0.023	+0.025	mixed
United Kingdom	1995–2023	+0.034	+0.016	+0.018	mixed
Japan	1980–2023	+0.061	+0.005	+0.056	scale
Korea	1975–2023	+0.119	−0.003	+0.122	scale
France	1978–2023	+0.029	−0.003	+0.032	scale

Channel contributions in percentage points of $\Delta\delta$. Where the two channels offset or $\Delta\delta$ is small (the Netherlands, Iceland, Sweden), the multiplicative shares are numerically unstable and are omitted.

Two attributions are firm and agree in direction with a PWT-based decomposition: the **United States and most advanced economies are composition-led** — the effective depreciation rate rose while the capital-output ratio was flat or fell — and **Japan is scale-led**, driven by capital deepening. These anchors are stable.

The intermediate cases are not. France, Germany, and Korea attribute to the opposite channel here than under a PWT decomposition, and the divergence is partly a valuation difference: the current-replacement-cost stock used here yields a different K/Y dynamic than the real stock underlying a PWT decomposition, the two differing by the relative price of capital. We therefore report the country-level attribution as qualitative and source-dependent: both channels operate, the two anchor cases are firm, and the middle of the distribution should be read as attribution-ambiguous rather than assigned a channel. Implication I3 — that both channels operate and both compress n — does not depend on which leads in any particular country.

The composition channel does not transmit to growth at low frequency.¹ We note this

¹Regressing low-frequency labor-productivity growth (ten-year long differences and HP-filtered trends) on the effective depreciation rate CFC/ K versus the capital-output ratio K/Y , with country fixed effects on the 30-country OECD panel, yields no coefficient distinguishable from zero once standard errors are computed by wild cluster bootstrap (appropriate given the small number of long-series clusters); the CFC/ K coefficient is not robust to the choice of low-frequency filter and changes sign across specifications ($p \approx 0.7$). The conventional cluster-robust t -statistic on CFC/ K in ten-year differences is large ($t \approx 3.8$), but this is the few-cluster artifact the bootstrap is designed to correct. We

Dashboard: δ = CFC/GDP (Direct Statistics) and Productivity
PWT 11.0 · 180 Countries · 1955-2023

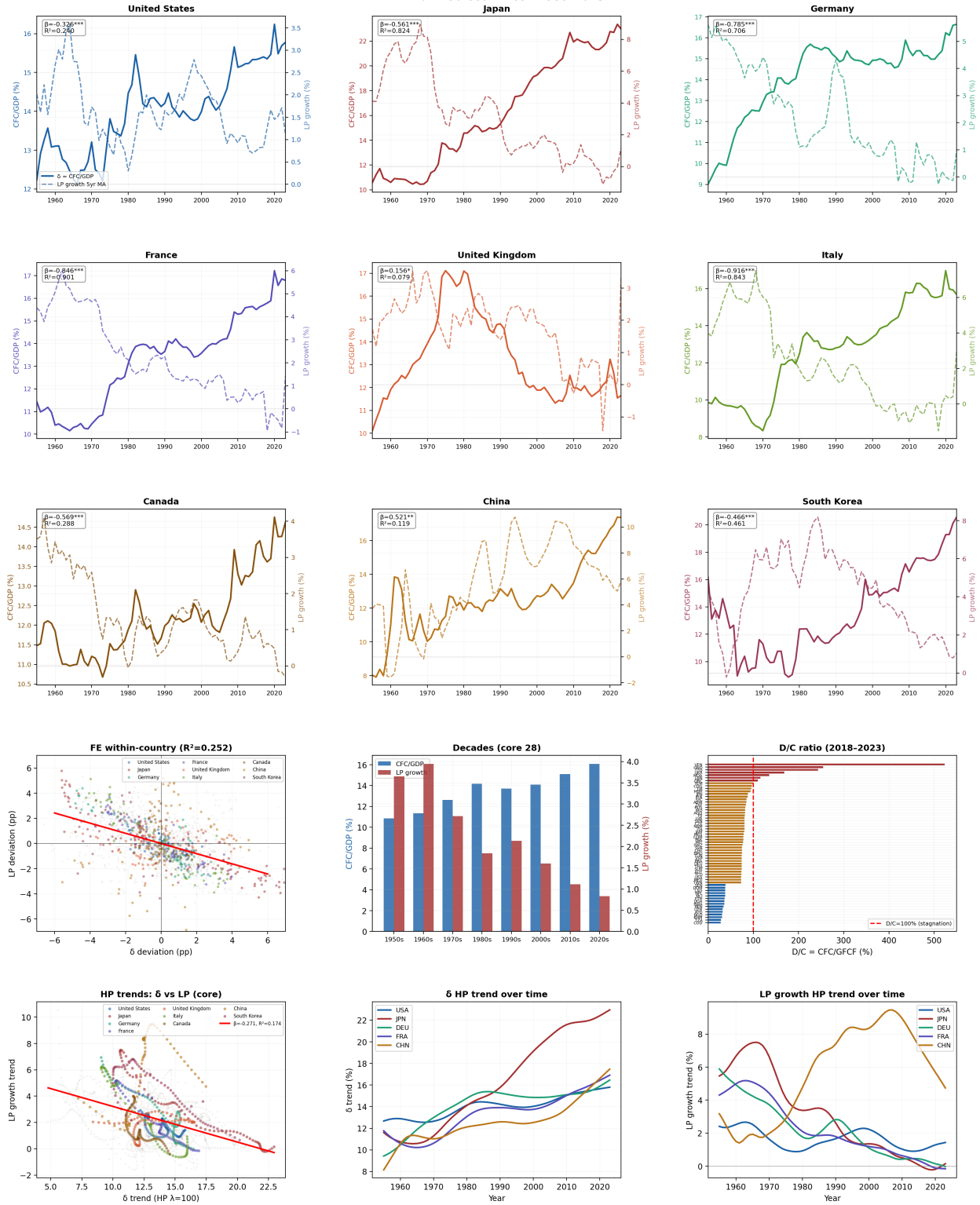


Figure 3: PWT dashboard

to fix the paper's boundary: the decomposition is an account of *how* δ rose, not a claim that either channel slows growth.

6. Layer 3: OECD STAN Industry Data

This layer uses $\delta = \text{CFC}/\text{VA}$, since GDP is not defined for individual sectors; industry-level δ can exceed aggregate δ in capital-intensive sectors, so we compare signs and qualitative patterns across layers, not point estimates.

6.1 δ rises in every country

δ rises in every STAN country, with trend slopes from 0.48 pp/decade (Australia) to 3.20 (Korea); all country R^2 exceed 0.50.

6.2 Net building compresses toward zero

Figure 4: D/C ratios approaching the stagnation bound (OECD STAN)

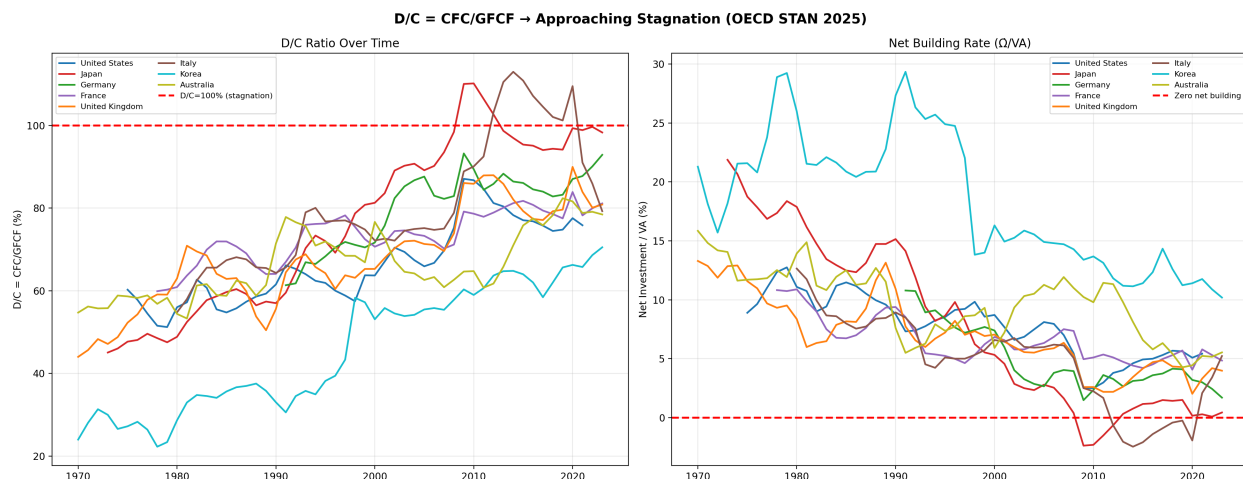


Figure 4: DC stagnation

Every country trends toward $D/C = 100\%$. Italy briefly breached it around 2012; Japan has held above 90% for over a decade. As an accounting check on I2 and I4, regressing the net investment rate on δ at the industry level gives a fixed-effects coefficient of -1.05 (SE 0.084, $N = 1,280$): rising δ compresses net investment, as $n = C/Y - \delta$ requires.²

read this as confirming that the composition/scale decomposition is an accounting description, and that identifying any transmission to growth requires variation we do not have here.

²The coefficient exceeds unity in absolute value because it captures covariance between GFCF/VA and CFC/VA beyond the mechanical identity. This is a regression on n , the dependent variable at which the paper stops, not a productivity regression.

7. The Limiting Case

7.1 Japan: the replacement trap in practice

Figure 5: Japan — D/C approaching 100% and the collapse of net investment

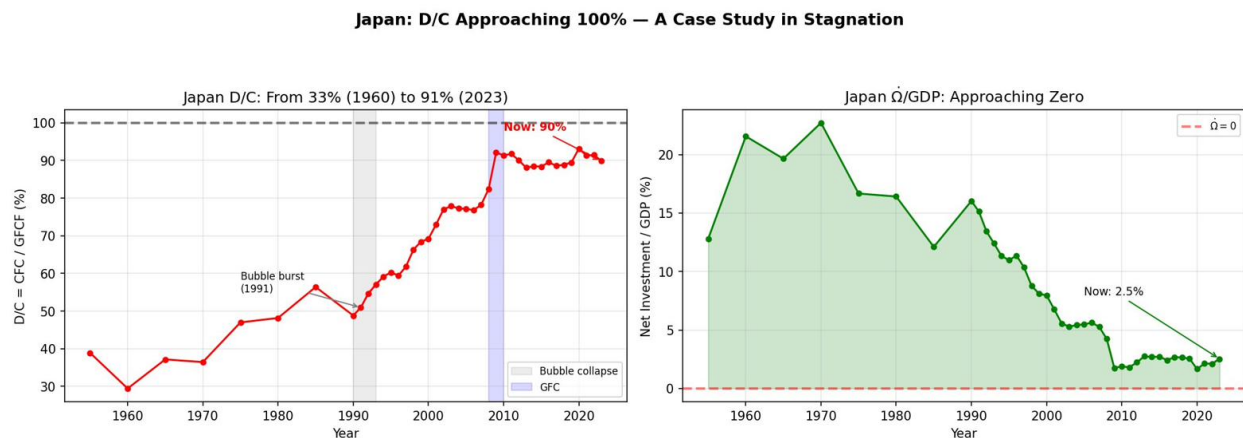


Figure 5: Japan

Japan's D/C rose from 33% in 1960 to 91% in 2023; net investment fell from 22% of GDP in 1970 to 2.5% in 2023. With 91% of gross investment absorbed by replacement, 9% remains for net new building — the closest empirical approach to the replacement trap of Section 2.2. We do not claim Japan proves the trap: its lost decades are also attributed to demand-side factors (Koo 2011) and banking dysfunction (Caballero, Hoshi, and Kashyap 2008), and many forces operated at once. Japan is distinguished from other economies not in kind but in degree and duration.

7.2 Cross-country trajectories

Figure 6: D/C ratios, international comparison

D/C ratios range from about 40% (China) to 91% (Japan). The ordering — China, US, Germany, Japan — coincides with the ordering of net building rates: economies further along the δ trajectory retain less room for net new building.

8. Discussion

8.1 A consistent reading of the slowdown literature

The diverse strands of the slowdown literature may describe different *sources* of one accounting fact rather than competing causal accounts. Each named force raises δ through a different sub-component, and each is positively associated with δ . On this reading the R&D productivity paradox is reinterpreted rather than resolved: R&D creates fast-depreciating capital, and the replacement cost of what it creates offsets part of its measured benefit. We offer this as a consistent reading; establishing it as a causal decomposition is beyond what reduced-form evidence supports.

D/C Ratio: International Comparison (US, Japan, Germany, China)
D = Consumption of Fixed Capital, C = Gross Fixed Capital Formation

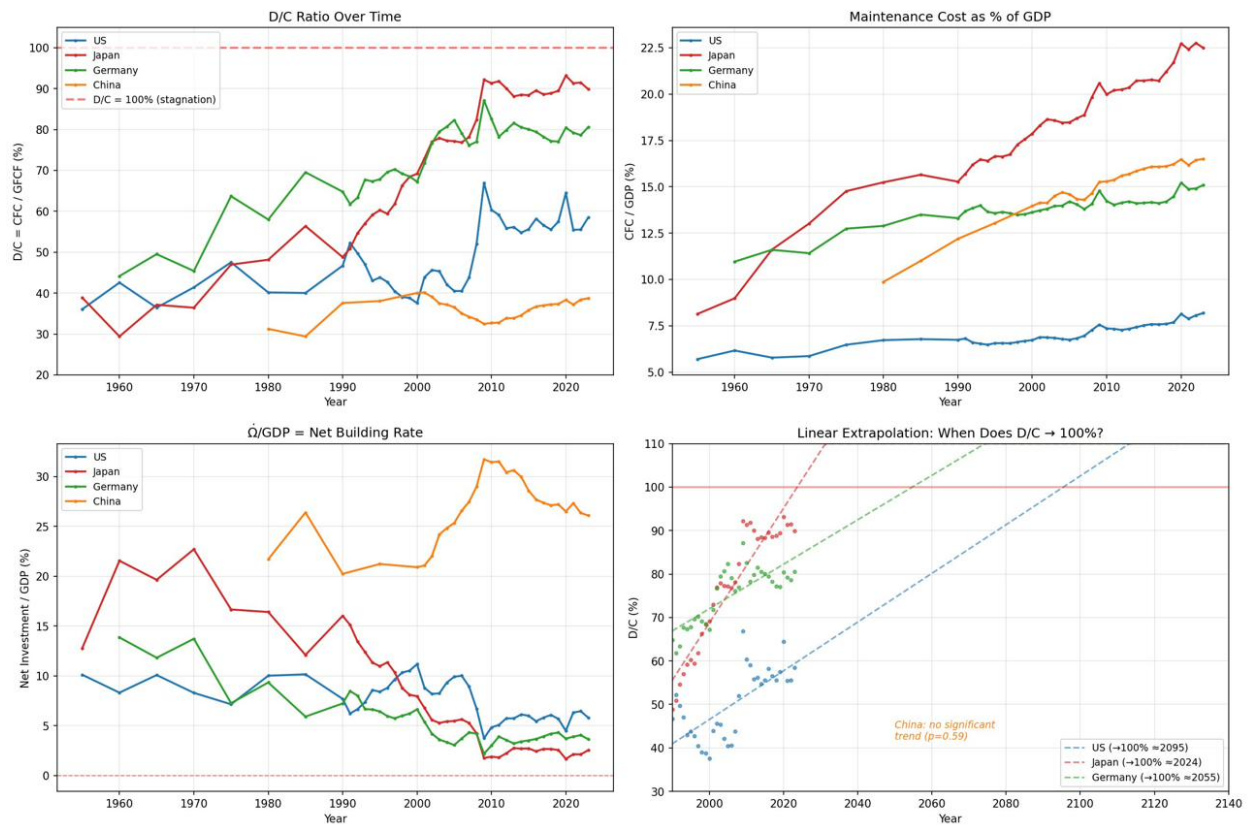


Figure 6: International DC

8.2 What we do not claim, and why

It is tempting to convert the δ -productivity co-movement into a statement about for-gone growth. We decline, for three reasons. The relationship between net accumulation and productivity has a debated structural magnitude, on which we take no stand. In observational data the variation that would identify a causal channel from δ to growth is difficult to separate from reverse causation and from movements in the shared output denominator. And our own low-frequency tests do not identify a transmission through the composition channel (Section 5.3, note). The honest output is a measurement and an identity: δ rises, decomposes, and compresses n . Whether the compressed n slows growth, and by how much, is deferred to work that can isolate the relevant variation.

8.3 Policy implications, conditional

These follow only if a causal channel from the replacement burden to growth is later established. If the slowdown is substantially a replacement-cost problem, demand-side intervention would not reverse it, and the relevant margin would be the *composition* of investment — a shift toward durable, low-maintenance assets. Whether AI eases or worsens the burden would turn on whether it reduces D (automating maintenance) or raises it (creating short-lived digital infrastructure). We state these as conditional implications, not recommendations.

9. Conclusion

The share of output absorbed by replacing depreciating productive structure has risen across every major economy for seven decades. The rise is physical, invariant to accounting method, and present at three levels of aggregation. Because the gross investment share is bounded, it tightens an accounting constraint on net new building — a constraint that, in Japan, has compressed net capital formation to 2.5% of GDP. It decomposes into a composition channel and a scale channel; the United States is composition-led, Japan scale-led, and the intermediate attribution is sensitive to valuation and best read qualitatively.

The paper rests on an accounting identity and a single monotonicity assumption, and we have been deliberate about what the evidence supports: a measured rise, a decomposition, and a compressed residual — not a causal effect on productivity growth. Whether the compressed net building rate slows growth runs through a transmission coefficient we do not identify, and our own low-frequency tests do not recover it. Isolating that channel — separating the replacement-burden signal from the reverse causation and denominator movements that travel with it — is the task we take up next.

Data and Code Availability

All data and replication code are available at <https://doi.org/10.17605/OSF.IO/GPKZN>. The package reproduces all results across the three layers, including the low-frequency transmission tests of the Section 5.3 note, and runs an automated validation script.

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(Appendix E references — Goldsmith-Pinkham, Sorkin, and Swift 2020; Conley, Hansen, and Rossi 2012; Oster 2019; Arkhangelsky et al. 2021; Borusyak, Hull, and Jaravel 2022 — retained in the full reference list.)

Appendices

All appendix material is provided in the companion **Online Appendix**: data construction and technical details (A), the illustrative two-sector model (B, qualitative only), German reunification as a historical illustration (C), the robustness summary (D), and — separately and explicitly *not* part of this paper’s claims — material toward causal identification (E), which collects the shift-share instrument, the natural experiments, the productivity-on- δ regressions, and the low-frequency composition test reported as a negative result in the Section 5.3 note.